

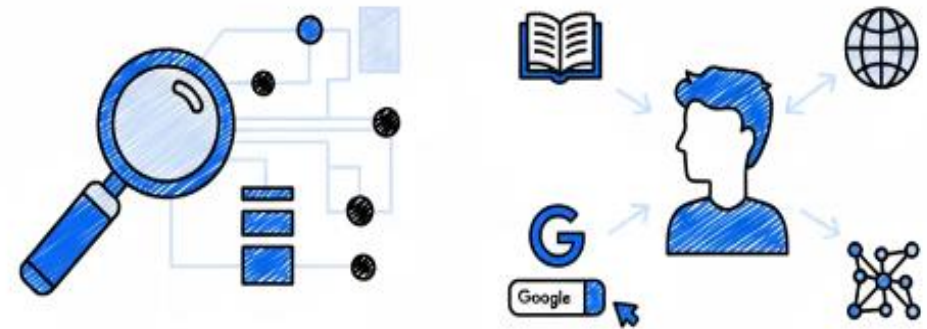
Red Team Recon

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- Group: 5
- Student:6
- Topic: Red Team Recon

Red Team Recon

- The material focuses on passive reconnaissance—techniques that do not alert the target or create 'noise'.
- Passive recon relies on publicly available information collected by a third party (Open-Source Intelligence or OSINT).

Passive Reconnaissance



Whois

- A protocol to query public database for the registration information of domain name
- Registrar server/URL, Record creation date,
- Record update date, and contact info (Registrant, Admin, Tech), unless withheld for privacy.

```
root@ip-10-10-232-159:~# dig clinic.thmredteam.com A +short | wc -l
2
root@ip-10-10-232-159:~# dig clinic.thmredteam.com AAAA +short | wc -l
2
root@ip-10-10-232-159:~#
```

```
eseller: NAMECHEAP INC
omain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
egistry Registrant ID:
egistrant Name: Redacted for Privacy
egistrant Organization: Privacy service provided by Withheld for Privacy ehf
egistrant Street: Kalkofnsvegur 2
egistrant City: Reykjavik
egistrant State/Province: Capital Region
egistrant Postal Code: 101
egistrant Country: IS
egistrant Phone: +354.4212434
egistrant Phone Ext:
egistrant Fax:
egistrant Fax Ext:
egistrant Email: e17b7976233e4e72a76b3dadb1d574bd.protect@withheldforprivacy.co

egistry Admin ID:
dmin Name: Redacted for Privacy
dmin Organization: Privacy service provided by Withheld for Privacy ehf
dmin Street: Kalkofnsvegur 2
dmin City: Reykjavik
dmin State/Province: Capital Region
dmin Postal Code: 101
```

```
root@ip-10-10-141-20:~# whois thmredteam.com
Domain Name: THMREDTEAM.COM
Registry Domain ID: 2643258257_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.namecheap.com
Registrar URL: http://www.namecheap.com
Updated Date: 2025-08-25T08:54:53Z
Creation Date: 2021-09-24T14:04:16Z
Registry Expiry Date: 2026-09-24T14:04:16Z
Registrar: NameCheap, Inc.
Registrar IANA ID: 1068
Registrar Abuse Contact Email: abuse@namecheap.com
Registrar Abuse Contact Phone: +1.6613102107
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransf
rohibited
Name Server: KIP.NS.CLOUDFLARE.COM
Name Server: UMA.NS.CLOUDFLARE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wi
>>> Last update of whois database: 2025-11-07T20:33:02Z <<<

For more information on Whois status codes, please visit https://icann.org/ep
NOTICE: The expiration date displayed in this record is the date the
```

- WHOIS and DNS-based recon, Advanced searching, Google Hacking, Specialized search engines, and tools like Recon-ng and Maltego uses to gather information about the target
- The Whois command shows how to collect information about the target

DNS and traceroute

- Queries DNS servers to resolve domain names.
- Nslookup, dig or host can be used to query DNS server as you can see on the pic
- Traceroute is a tool that map the digital pathway or hops data takes to reach a target destination.
- Used to discover the **hops** between the attacker's system and the target host.

```
root@ip-10-10-232-159:~# dig clinic.thmredteam.com A +short | wc -l
2
root@ip-10-10-232-159:~# dig clinic.thmredteam.com AAAA +short | wc -l
2
root@ip-10-10-232-159:~#
```

Reverse IP Lookup
Find all domains hosted on a given IP address.

Enter Domain or IP address

HTML JSON

Reverse IP results for cafe.thmredteam.com (104.21.93.169, 172.67.212.249)
There are 1,033 domains hosted on this server.
The first 500 of these are listed below.

DOMAIN NAME	LAST RESOLVED
138ei.com	2021-05-10
138wv.com	2021-05-10
1767555.app	2025-11-01
17tg8.cn	2025-11-05

For more information on Whois status codes, please visit <https://ica>

```
root@ip-10-10-141-20:~# nslookup cafe.thmredteam.com
Server:      ::1
Address:     ::1#53

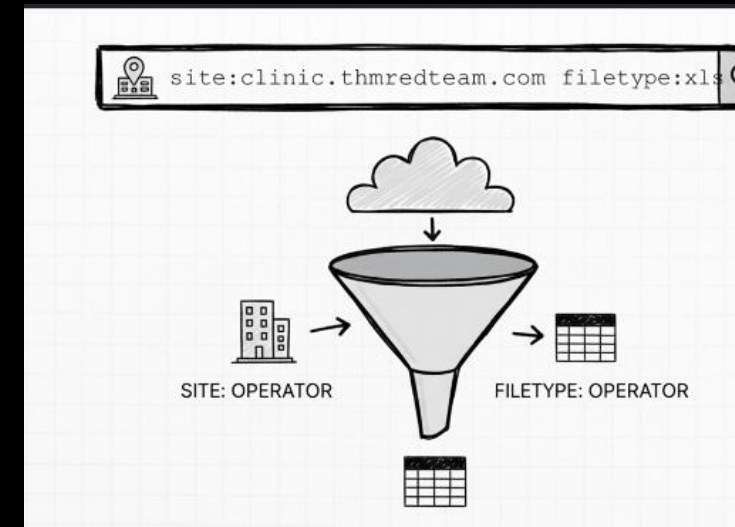
Non-authoritative answer:
Name:   cafe.thmredteam.com
Address: 172.67.212.249
Name:   cafe.thmredteam.com
Address: 104.21.93.169
Name:   cafe.thmredteam.com
Address: 2606:4700:3034::ac43:d4f9
Name:   cafe.thmredteam.com
Address: 2606:4700:3034::6815:5da9

root@ip-10-10-141-20:~#
```

Advanced Searching (Google Hacking / Dorking)

- Search engines frequently index confidential information unintentionally.
- Confidential Information Exposed:
 - Filetype:xls site:clinic.thmredteam.com searching can find file as you can see in the pic
 - Documents intended for internal company use
 - Confidential spreadsheets containing usernames, email addresses, and even passwords.

	First	Last	Degree(s)	Profession	Country	State	City	Phone	Email	Website	Yrs in Practice	Yrs in CBT	CISM Certified
1	H.	Mitchell	EdD	Clinical Psychologist	USA	AL	Birmingham	205-879-0202	hdmitchell@gmail.com	www.dicomitchell.net	30	4	Planned
2	William	Bryson	PHD	Clinical Psychologist	USA	AL	Huntsville	(256) 533-9393					
3	Kenneth	Uchstein	PHD, CISM	Clinical Psychologist	USA	AL	Tuscaloosa	(205) 348-4962					
4	Arlene	Sullivan	MNSc, APRN	Nurse Practitioner	USA	AR	Little Rock	501-202-1902	arsullivan@practice-plus.com		12	8	N
5	Robert	Boyle	PHD, RN, CT, NCC, AHN, BC, FAAN	Clinical Health Psychologist	USA	AR	Sherwood	501-771-4442	ARBMS@ARBMS.org	www.ARBMS.com	43	35	N
6	Dana	Epstein		Nurse	USA	AZ	Phoenix	602-277-5551 ext 37725	dana.epstein@va.gov		21	21	N
7	Russell	Gilbert	MD, CISM, PHD, RN, CT, NCC, AHN, BC, FAAN	Physician	USA	AZ	Scottsdale	(480) 467-0300					
8	Carol	Baldwin		Psychologist/Counselor	USA	AZ	Tucson		carol.baldwin@asu.edu		15	0	N
9	Tricia	Haynes	PHD	Clinical psychologist	USA	AZ	Tucson	520-626-1855			9	9	



Recon-ng

The command **recon-ng -w thmredteam** start recon-ng with the workspace as the pic shows

To find the modules with the name virustotal run the command -> marketplace search virustotal

```
[recon-ng][clinicredteam] > db insert domains
domain (TEXT): thmredteam.com
notes (TEXT):
1 rows affected.
[recon-ng][clinicredteam] > marketplace search
```

+-----+-----+-----+-----+-----+-----+					
Updated		D	K	Path	Status
+-----+-----+-----+-----+-----+-----+					
2020-10-13				discovery/info_disclosure/cache_snoop	not installed
2021-10-04				discovery/info_disclosure/interesting_files	not installed
2019-06-24				exploitation/injection/command_injector	not installed

```
[recon-ng][thmredteam] > marketplace search virustotal
Searching module index for 'virustotal'...
```

+-----+-----+-----+-----+-----+-----+					
Updated		D	K	Path	Status
+-----+-----+-----+-----+-----+-----+					
2019-06-24				recon/hosts-hosts/virustotal	not installed
2019-06-24				recon/netblocks-hosts/virustotal	not installed

D = Has dependencies. See info for details.
K = Requires keys. See info for details.

Recon-ng

Searches installed modules

```
Usage: modules search [<regex>]
```

```
[recon-ng][default] > marketplace search hosts-domains
Searching module index for 'hosts-domains'...
```

D = Has dependencies. See info for details.
K = Requires keys. See info for details.

```
[recon-ng][default] >
```

```
[recon-ng][default] > marketplace info censys_email_address
```


Recon-ng

```
[recon-ng][default] > marketplace info censys_email_address
```

```
+-----+
+-----+
+-----+
| path          | recon/companies-contacts/censys_email_address |
| name          | Censys - Emails by Company                    |
| author        | Censys, Inc. <support@censys.io>              |
| version       | 2.1                                             |
| last_updated  | 2022-01-31                                     |
| description   | Retrieves email addresses from the TLS certificates for a co
```

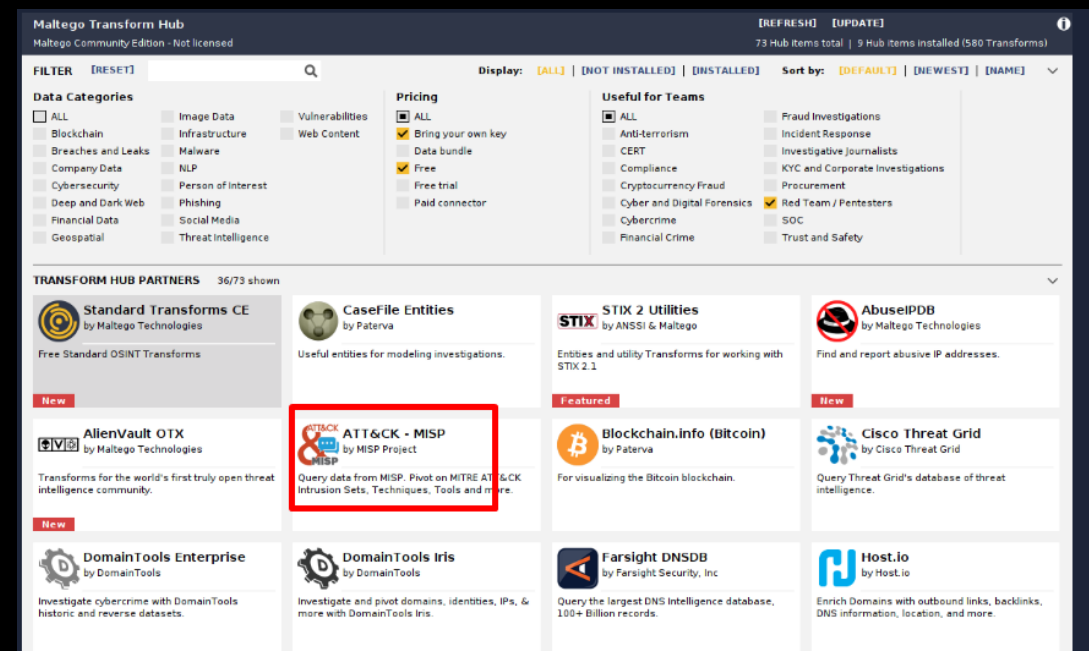
Social Media



- Platforms like LinkedIn, Twitter, Facebook, and Instagram can reveal significant information due to users oversharing details about themselves and their work.
- Information gathered can include employee names and specific pieces of information that reveal answers to password recovery questions or help build targeted wordlists.
- Posts from technical staff might inadvertently reveal details about company systems

Maltego

- Building blocks (domains, IPs, emails, people)
 - Code that queries APIs to discover related information
 - Each transform can reveal multiple new entities
 - DNS Name (cafe.thmredteam.com)->"To IP Address" Transform ->IP Address Entities -> Additional Transforms
 - the transform that queries NIST's National Vulnerability Database is s
- NIST NVD
- MISP project offers a transform based on ATT&CK



Observations of seminar sessions

- During the lab session, I found that the material was largely familiar, as I had already studied all the topics before to the seminar, and the presentations by my peers introduces the same concepts. However, the method by which the malicious file downloaded to the VM's desktop and executed in the terminal to locate the hash file was new to me.
- Previously known techniques that were validated include the use of WHOIS and DNS-based reconnaissance, advanced searching techniques, Google hacking, specialized search engines, and programs such as Recon-ng.
- I did not observe anything that I believed I understood but was afterwards proven incorrect.



Thanks